Katy Felkner

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Research Summary

Final year Ph.D. candidate graduating April 2027 and seeking **full-time roles beginning June 2027.** Research agenda focuses on developing high-quality, **well-grounded benchmark datasets** for robust evaluation of language models, with particular emphases on participatory, community-engaged methods and on rigorous measurement of fairness and social bias.

Education

University of Southern California — Los Angeles, CA
Ph.D. in Computer Science — Aug. 2020 – Present

- Expected Apr. 2027; GPA: 3.90/4.00; Supported by **NSF Graduate Research Fellowship**
- Coursework: Advanced Natural Language Processing, Fairness in AI, Probability and Statistics for Data Science, Representation Learning in NLP, Advanced Algorithms, Crafting a Research Agenda in Computer Science
- Advised by Jonathan May

University of Oklahoma — Norman, OK
B.S. in Computer Science (summa cum laude), B.A. in Letters (summa cum laude) — Aug. 2016 – May 2020

- Minor: Mathematics; GPA: 4.00/4.00

Publications

Felkner, Virginia K., Allison Lim, and Jonathan May (Dec. 2026). "Textual Entailment is not a Better Bias Metric than Token Probability". In: *QueerInAI/{Dis}AbilityInAI Workshop*. NeurIPS 2026. arXiv: 2510.07662 [cs.CL].

Felkner, Virginia K., Jennifer Thompson, and Jonathan May (Aug. 2024). "GPT is Not an Annotator: The Necessity of Human Annotation in Fairness Benchmark Construction". In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*. DOI: 10.18653/v1/2024.acl-long.760.

Felkner, Virginia K., Ho-Chun Herbert Chang, et al. (July 2023). "WinoQueer: A Community-in-the-Loop Benchmark for Anti-LGBTQ+ Bias in Large Language Models". In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*. DOI: 10.18653/v1/2023.acl-long.507.

Felkner, Virginia K., Ho-Chun Herbert Chang, et al. (July 2022). "Towards WinoQueer: Developing a Benchmark for Anti-Queer Bias in Large Language Models". In: *QueerInAI Affinity Group Workshop*. NAACL 2022. arXiv: 2206.11484 [cs.CL]. URL: https://arxiv.org/abs/2206.11484.

Weideman, Nicolaas, **Virginia K. Felkner**, et al. (Nov. 2021). "PERFUME: Programmatic Extraction and Refinement for Usability of Mathematical Expression". In: *Proceedings of the 2021 Research on Offensive and Defensive Techniques in the Context of Man At The End (MATE) Attacks*. ACM SIGSAC CCS 2021. DOI: 10.1145/3465413.3488575.

Felkner, Virginia K. and Elisabeth (Lissa) Moore (Nov. 2020). "Investigating the Efficacy of Unstructured Text Analysis for Node Failure Detection in Syslog". In: *Second Workshop on Machine Learning for Computing Systems*. Supercomputing 2020. URL: https://mlcsworkshop.weebly.com/uploads/1/1/6/9/116940258/ws_mlcs103s2-file1.pdf.

Research Experience

MIT Lincoln Laboratory — Boston, MA
Summer Research Program Intern — May 2026 – present

- Led a study of demographic performance disparities in LLM-based relation extraction; independently defined the fairness framework, demographic taxonomy, and evaluation design.
- Constructed a demographic annotation layer over 1,700 unique KnowledgeNet/Wikidata entities.
- Ran large-scale, multi-node evaluations across 8 models (up to 400B parameters) and 16 prompt conditions.
- Applied two-proportion z-tests, chi-square tests, and Fisher's exact tests to detect performance disparities; identified statistically significant effects across gender and geographic axes.
- Delivered 2 technical presentations to internal team and government sponsor; both received strong commendation.
- Continuing as part-time Student Technical Assistant after end of internship to extend analysis to larger FewRel dataset; targeting submission to a top NLP venue via ARR (January 2027 deadline).

USC Information Sciences Institute — Los Angeles, CA
Graduate Research Assistant – LLM Fairness and Bias — Jan. 2022 – Present

- Develop two community-sourced benchmark datasets for detecting anti-LGBTQ+ and antisemitic bias in LLMs.
- Design and distribute two community surveys and analyze 500 responses to inform LLM bias benchmarks.
- Lead team of 3 to collect 8M sentences of community data, then train 32 debiased models via continued pretraining.

- Assess LLM capability to assist in fairness benchmark construction via detailed human analysis of model failures.
- Analyze viability of natural language inference (NLI) as an LLM bias metric, developing a novel NLI bias dataset.
- **Work in Progress:** Assess demographic fairness in realistic résumé evaluation task.
- **Publications:** Lead interdisciplinary teams to publish four papers (two ACL, two workshopf).
- **Mentoring:** Mentored undergraduate student on NLI bias metric project, Jan. – Oct. 2025. Currently mentoring masters' student on project relating to LM sycophancy caused by emotional language in prompts.

Graduate Research Assistant – DARPA ReMATH Jan. 2021 – Jun. 2022

- Trained neural machine translation (NMT) models for extraction of symbolic math from bit-vector expressions.
- Developed novel metrics for evaluation of NMT-generated mathematical expressions.
- **Publication:** Wrote relevant sections of workshop paper accepted to CheckMATE, ACM CCS 2021.

Graduate Research Assistant – Border Translation Aug. 2020 – Dec. 2021

- Built and evaluated NMT systems for extremely low-resource languages.
- Developed baseline NMT system for K'iche', a Mayan indigenous language, with intended application of helping NGOs provide legal assistance to immigrants and asylum seekers.

Ultrascale Systems Research Center, Los Alamos National Laboratory Remote

Post-Baccalaureate Research Intern May 2020 – Aug. 2020

- Collected 750,000 lines of non-secret syslog data using a 20-node virtual compute cluster.
- Tested several embedding and clustering methods to extract natural language features from supercomputer syslog.
- Found these natural language features can be used to improve HPC node failure detection and prediction.
- **Publication:** Authored workshop paper accepted to Second Workshop on Machine Learning for Computing Systems, Supercomputing 2020.

Digital Latin Library, University of Oklahoma Norman, OK

Undergraduate Research Assistant Jan. 2017 – May 2020

- Wrote Python scripts to parse HTML documents containing Latin texts, generating 125000 lines of SQL output.
- Designed scripts to automate XML encoding of Latin prose and poetry, reducing time to encode by a factor of 400.
- Only undergraduate student accepted to National Endowment for the Humanities Institute on Advanced Topics in Digital Humanities for Summer 2018 at Tufts University.
- **Presentations:** Conference talks at Society for Classical Studies 2019, Classical Association of the Middle West and South 2018, National Conference on Undergraduate Research 2018

Epic Systems Corporation Madison, WI

Software Developer Intern May 2019 – Aug. 2019

- Prototyped a menstrual cycle tracking feature for iOS MyChart to allow patients to share data with providers.
- Designed and implemented server and client-side data structures to save cycle data to patient's medical record.
- Built four complete and interactive user-interface screens and partial prototypes of several others.
- Conducted user research, including about 100 internal survey responses and 5 user interviews.

Technical Skills

Programming Languages and Skills: Python, C, C++, R, SQL, Bash Scripting, HPC, Slurm

ML/NLP/Data Science Tools: HuggingFace, PyTorch, TensorFlow, SpaCy, Pandas, Numpy, Scikit-Learn, MatPlotLib

Teaching Experience

University of Southern California Los Angeles, CA

Teaching Assistant – DSCI 510 Principles of Programming for Data Science Jan. 2026 – May 2026

- Prepared and led weekly lab sessions on beginner Python, git, shell, and data science topics.
- Graded 14 weekly programming assignments and one midterm exam for 40 students.
- **Student Feedback:** 80% "excellent" overall TA rating.

Teaching Assistant – CSCI 662 Advanced Natural Language Processing Aug. 2025 – Dec. 2025

- Updated homework assignments to better cover finetuning, attention, transformers, and RAG.
- Migrated automated grading to Gradescope and configured all assignments and rubrics.
- Graded assignments, including programming homework, research reports, and in-class presentations.
- Managed course logistics for about 30 students, including webpage, Slack channel, and Brightspace LMS.
- **Student Feedback:** 85% "excellent" overall TA rating.

Presentations

Invited Talks

- Celebrate Track Invited Speaker, *Community-Sourced Bias Metrics for LLMs*, UCLA QWERHacks. *Feb. 2025*
- Guest Lecturer on *Ethics and Power in NLP*, USC CSCI 662 Advanced NLP *Fall 2023, 2024, 2025*
- Invited Panelist on *Language Matters: Addressing Bias in LLMs*, IAPP Privacy. Security. Risk. *Sep. 2024*
- Invited Speaker, John Brooks Slaughter Leadership in Engineering DEI Summit. *Feb. 2024*
- Invited Panelist on *Bias in Generative AI*, QueerInAI Workshop at NeurIPS. *Dec. 2023*
- Invited Speaker, *Anti-Queer Bias in LLMs*, USC ISI Natural Language Seminar. *Jun. 2022*
- Guest Lecturer, *Servius: An Object-Oriented Approach*, Sunoikisis Digital Classics *May 2019*

Conference Talks and Posters

- Lightning Talk, ACM AI Leadership Summit Doctoral Consortium (selective; fully funded). *Aug. 2026*
- Lightning Talk, *Participatory Methods for LLM Bias Benchmarks*, Responsible Data Workshop. *Sep. 2025*
- Poster, *Textual Entailment as a Bias Evaluation Metric for LLMs*, USC ShowCAIS. *Apr. 2025*
- Poster, *GPT is Not an Annotator*, Southern California NLP Symposium. *Nov. 2024*
- Roundtable Discussion, *Defining and Disrupting Antisemitism*, Association for Jewish Studies. *Dec. 2023*
- Poster, *Anti-Queer Bias in LLMs*, CRA-WP Grad Cohort for Women (selective; fully funded). *Apr. 2023*
- Poster, *Anti-Queer Bias in LLMs*, Southern California NLP Symposium. *Nov. 2022*
- *Automatically Encoding Digital Editions of Latin Texts*, Society for Classical Studies. *Jan. 2019*
- *Coding And/As Classical Scholarship*, Classical Association of the Midwest and South. *Apr. 2018*
- *Coding And/As Classical Scholarship*, National Conference on Undergraduate Research. *Apr. 2018*

Professional Service

USC ISI Natural Language Seminar Los Angeles, CA
Coordinator Aug. 2024 – May 2025

- Recruited external speakers and coordinated publicity for 15 talks, averaging 20 attendees each.

Conference and Workshop Reviewing

- **Conferences:** EMNLP 2023, ACL 2023, EMNLP 2022 Theme Track
- **Workshops:** NAACL Student Research Workshop 2025
- **Ethics Reviewing:** EMNLP 2025, COLM 2026

Leadership and Outreach

USC Center for AI in Society Los Angeles, CA
Student Leader Aug. 2025 – Present

- Assist with publicizing seminars, mentoring CAIS++ undergraduates, and planning ShowCAIS Symposium.

Queers in Engineering, Science, and Technology (QuEST) Los Angeles, CA
Graduate Representative Sep. 2020 – Present

- Recruit graduate students to QuEST and advocate for graduate student needs in the organization.
- Conduct general meetings on graduate school/fellowship applications and mentor undergraduate applicants.
- 2022 oSTEM Region F Conference Organizer: Present graduate application workshop; speak on Queer in Academia panel; organize and hoste charity livestream, raising $950 for LGBTQ+ community organizations in LA

Alpha Sigma Kappa – - Women in Technical Studies Norman, OK
President Apr. 2019 – Apr. 2020

- Oversaw 6 executive board members and served as chief risk management officer.
- Led chapter's transition to virtual operations due to COVID-19, rapidly establishing new programs to maintain sisterhood and encourage academic excellence during the pandemic.

Vice President Dec. 2017 – Dec. 2018

- Planned and executed nine recruitment events per semester, recruiting 32 new members.
- Mentored 9 officers on 5 committees, resulting in effective chapter social and new member education programs.

Out in Science, Technology, Engineering, and Math Norman, OK
Secretary Mar. 2017 – Dec. 2019

- Served on founding executive board of first oSTEM chapter in the state of Oklahoma.
- Planned and executed about two medium-scale events per month and managed event promotion and member communication.

Halliburton Women's Welcome Norman, OK
Program Committee Chair Aug. 2018, Aug. 2019

- Planned and executed a week-long program for 80+ incoming female students in engineering.
- Organized campus resource fair and inclusive professional dress workshop which received 91% positive participant feedback.